

# Improving relocalization accuracy via environment prior

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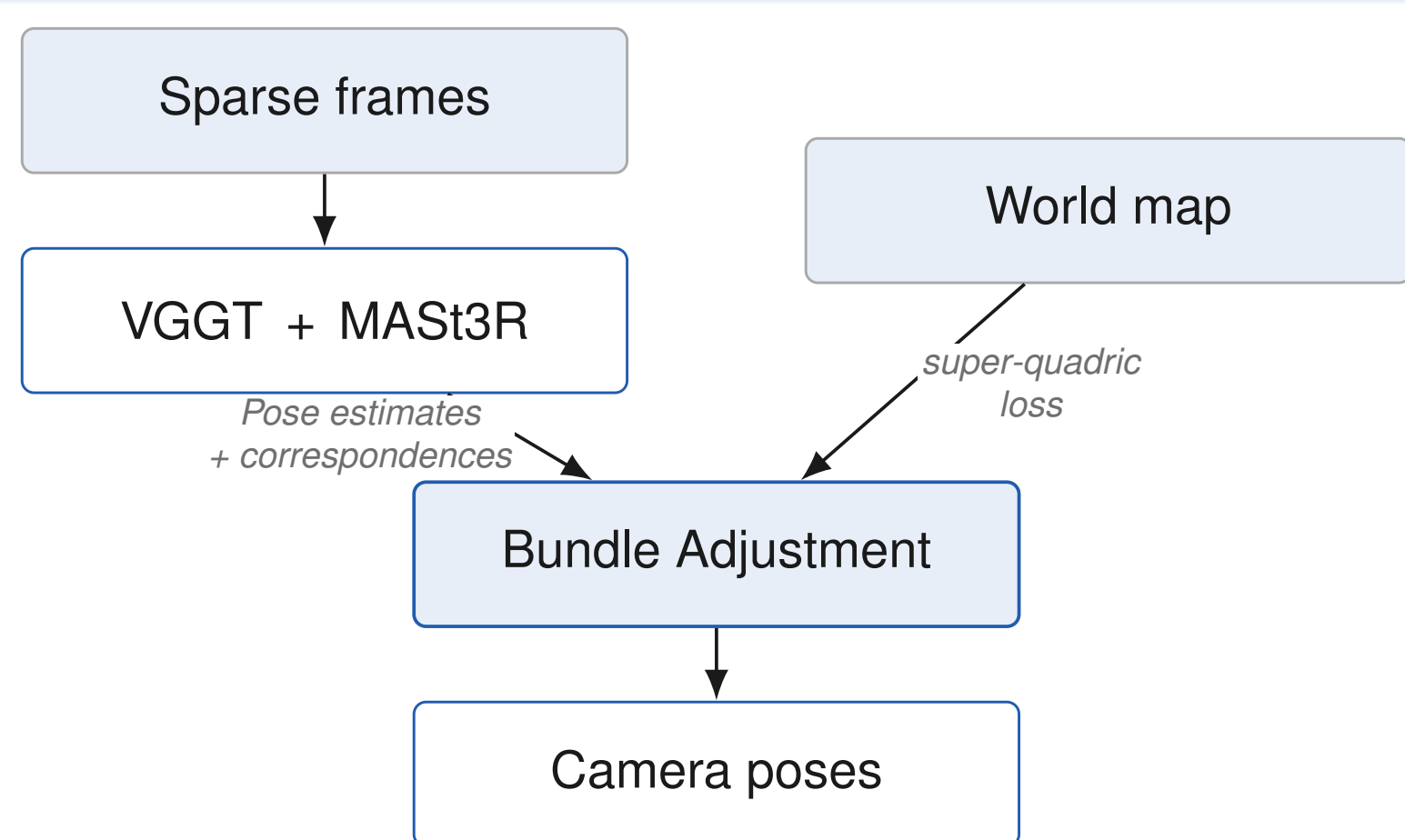
**A super-quadric environment prior corrects camera poses exactly where feed-forward SfM is weakest — sparse, low-overlap views.**

**+1.5 AUC@5° at 6 views (up to +2.0 at 4) — vs. only +0.2 at 10 views**

## 1 The Problem

In relocalization (the “kidnapped robot”), camera pose must be recovered accurately *given* a known environment. Transformer-based SfM (e.g. VGGT [1]) gives fast, strong feed-forward poses but **cannot exploit such a prior** — and it degrades when views are few and barely overlap. We bridge this gap with a **super-quadric** scene prior [4].

## 2 Method Overview



## 3 The Environment Prior

SuperDec [4] fits a compact set of **super-quadrics** to the scene point cloud. During BA, each triangulated point is pulled onto its nearest primitive by a one-sided point-to-quadric term, re-associated EM-style every few iterations:

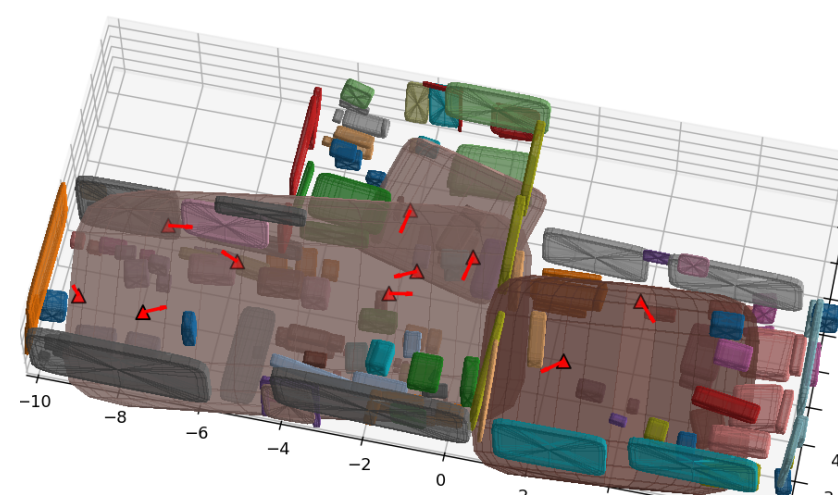


Figure 1: Super-quadric decomposition of a scene (189 primitives) with camera poses.

$$E = \underbrace{\sum_{i,j} \rho(\|\pi(R_i, t_i, X_j) - x_{ij}\|^2)}_{\text{reprojection}} + \underbrace{\sum_j \left( \lambda \|q_j\| \cdot \max(0, 1 - F(q_j)^{-\epsilon_1}) \right)^2}_{\text{one-sided surface prior}}$$

where  $F(\cdot)$  is the inside–outside function of the super-quadric nearest to point  $X_j$  and  $\lambda$  the prior weight [5].

## 4 Setup

- **Dataset:** Aria Synthetic Environments (10 scenes)
- **Models:** VGGT for feed-forward poses (via MapAnything [2]), MAST3R for matching [3]; SuperDec [4] for the prior
- **Optimization:** full triangulation + surface BA in Ceres
- **Metric:** pose AUC@5° (higher is better)

## References & Code

- [1] Wang et al., *VGGT: Visual Geometry Grounded Transformer*. CVPR 2025.  
 [2] Keetha et al., *MapAnything: Universal Feed-Forward Metric 3D Reconstruction*. 3DV 2026.  
 [3] Leroy et al., *Grounding Image Matching in 3D with MAST3R*. ECCV 2024.  
 [4] Fedele et al., *SuperDec: 3D Scene Decomposition with Superquadric Primitives*. ICCV 2025.  
 [5] Mueller et al., *Reconstructing People, Places, and Cameras*. CVPR 2025.



Code & report

## 5 Results

Three configurations: **VGGT-only** (raw feed-forward poses), **Baseline** (reprojection BA), and **Ours** (BA + the super-quadric prior).

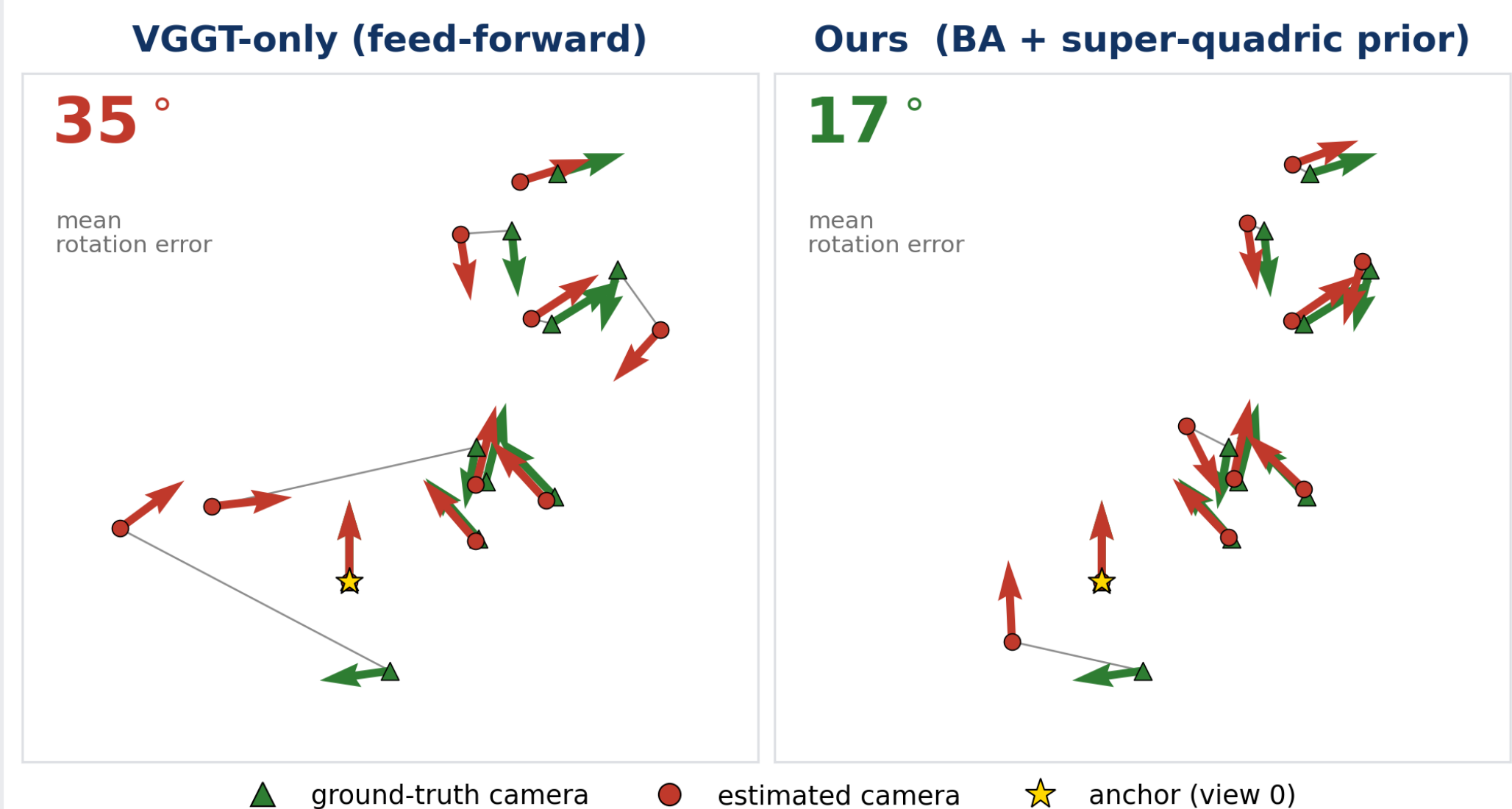
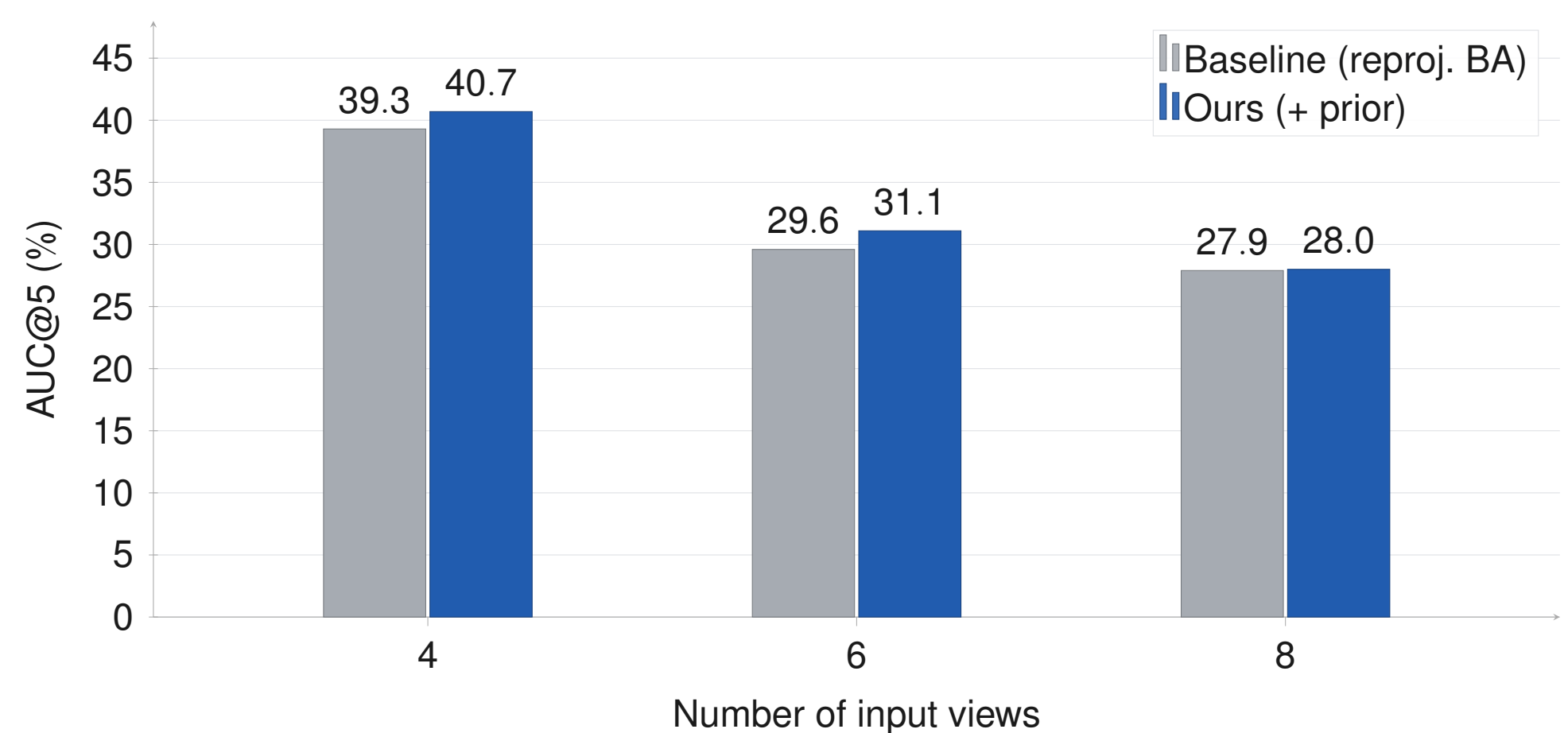


Figure 2: Our pipeline vs. raw VGGT on a scene — mean camera-rotation error drops 35° → 17° (estimated cameras snap onto ground truth).

### The prior's gain grows as views get sparser



| AUC@5 (%)   | # input views |             |             |             |
|-------------|---------------|-------------|-------------|-------------|
|             | 4             | 6           | 8           | 10          |
| VGGT-only   | 9.3           |             |             |             |
| Baseline    | 39.3          | 29.6        | 27.9        | 29.4        |
| <b>Ours</b> | <b>40.7</b>   | <b>31.1</b> | <b>28.0</b> | <b>29.6</b> |
| $\Delta$    | <b>+1.3</b>   | <b>+1.5</b> | +0.1        | +0.2        |

Up to +2.0 at 4 views with light  $\lambda$  tuning.

## 6 Conclusion

At high view counts dense correspondences already constrain the poses, so the prior is largely redundant (+0.2 at 10 views). As views drop and co-visibility collapses, reprojection alone under-constrains the geometry — and the super-quadric surface fills exactly that gap, recovering up to +2.0 AUC@5 at 4 views. **Encoding an environment prior as a super-quadric cloud helps relocalization most where it matters: sparse, low-covisibility views.**